



Fabric Architecture Review - Sample

FABRIC ARCHITECTURE REVIEW

Client: Contoso

Reviewer: Fabric Review Team

Review date: 2026-01-15

Tenant: ***

FAR version: 2026.09.0

Executive Summary

Sample report. All workspace, dataset, and tenant identifiers have been masked (*******) for public distribution. A real engagement run retains the actual per-resource IDs so every finding is directly actionable.

Scope

Metric	Value
Workspaces reviewed	3
Fabric capacities reviewed	2
Activity log lookback	7 days
Total checks executed	46
Failing checks	34
Passing checks	7
Informational	5
Not applicable	0
Unknown applicability	0
Missing evidence	0
Assessment coverage	100.0%

Findings overview

Dimension	Critical	High	Medium	Low	Info
Architecture	0	1	1	0	0
Performance	0	2	9	0	0
Governance	0	0	2	0	0
Operational Excellence	0	0	0	0	0
Security	1	2	1	0	0
Cost	0	0	2	0	0
Tenant Settings	0	1	0	0	0
Notebook Code Review (heuristic)	0	1	3	2	0

Counts above include **failing checks only**. Every outcome, including exclusions and evidence gaps, is listed in Detailed Findings for full traceability.

Top risks

Severity	Rule	Title	Impact
CRITICAL	SEC-001	Publish to web is disabled or restricted to a security group	Setting is enabled for the entire organization with no security group scoping.
HIGH	ARCH-011	1 lakehouse-backed Import model(s) may fit Direct Lake	—
HIGH	BPA-001	BPA-001: 2 model best-practice violation(s) across 2 model(s)	—
HIGH	BPA-002	BPA-002: 1 report issue(s) across 2 report(s)	—
HIGH	BPA-003	BPA-003: 1 Direct Lake fallback reason(s) detected	—

Severity	Rule	Title	Impact
HIGH	DAX-001	1 measure(s) contain potentially expensive static DAX patterns	—
HIGH	NBCODE-001	Notebook cells contain potential hard-coded secrets (1 cell(s) across 1 notebook(s))	—
HIGH	PERF-001	Throttling risk on 1 capacity(ies) (approaching rejection); worst P95 rejection 85%	—
HIGH	SEC-006	Misconfigured datasource instances	1 item(s) affected
HIGH	SEC-008	Single-member gateway cluster(s) bound to data sources (SPOF)	—

Methodology

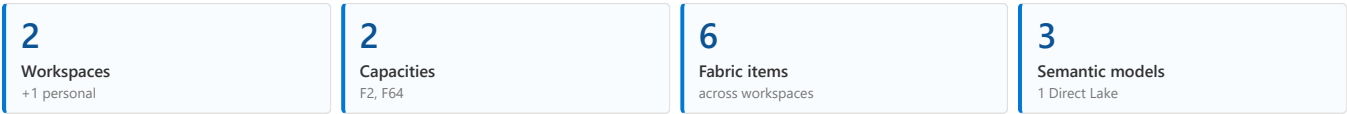
This review is aligned to the [Microsoft Azure Well-Architected Framework](#) and the [Fabric implementation planning](#) guidance. Checks run against tenant metadata, configuration and metrics surfaced through the Power BI and Fabric Admin REST APIs.

No customer data is read. Refer to the Data Safety appendix at the end of this report.

Environment Overview

An at-a-glance map of the Fabric estate captured by the collectors — the workspaces, items, capacities, access and activity this review is based on. Every number is metadata only; no customer data is read.

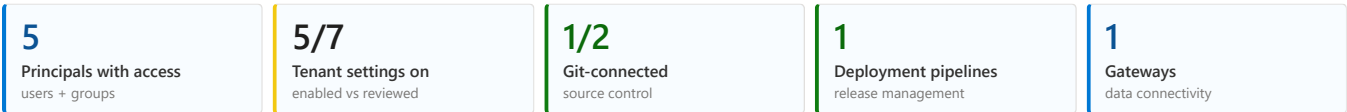
ESTATE



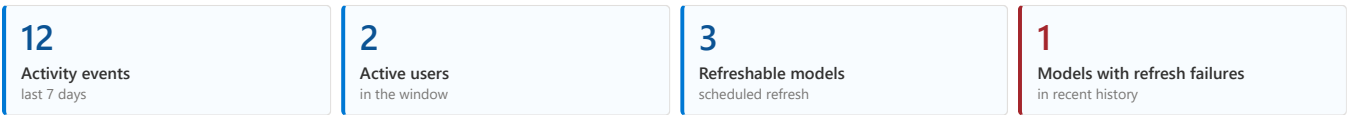
ITEMS BY TYPE



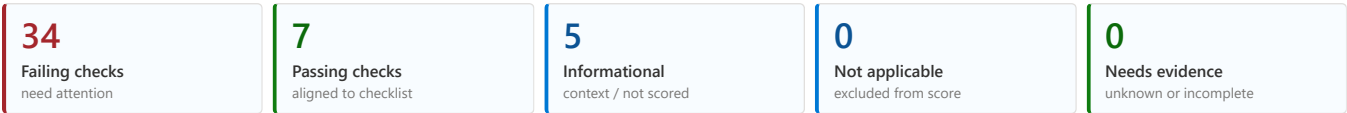
GOVERNANCE & ACCESS



ACTIVITY & REFRESH



REVIEW RESULT

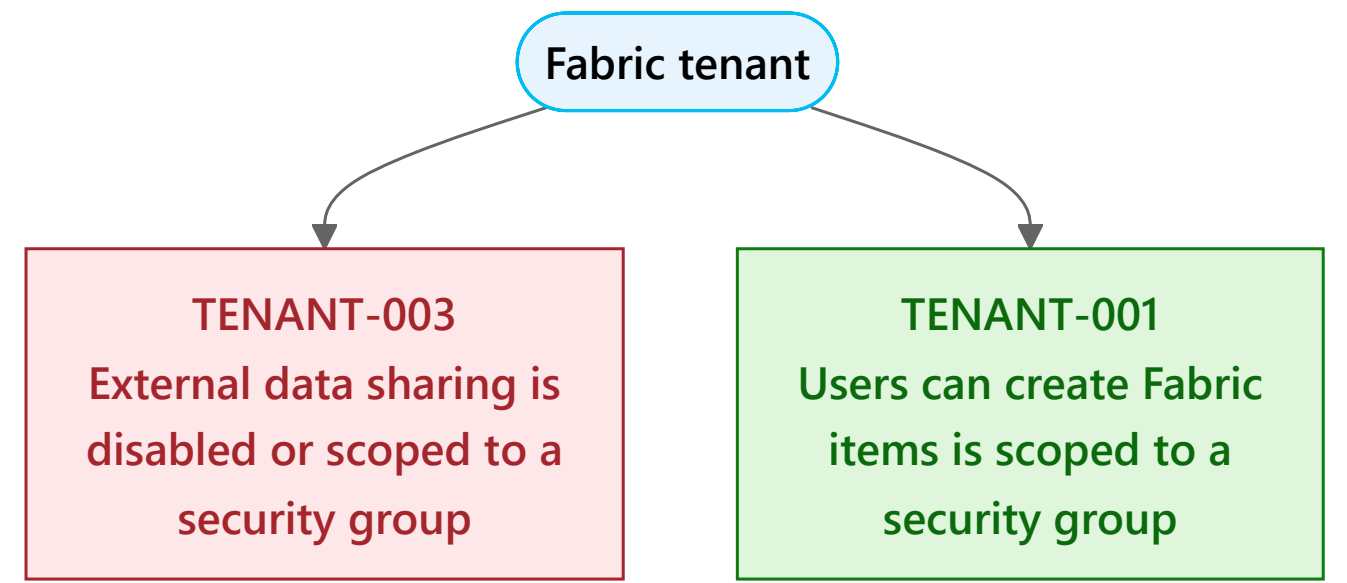


Architecture Overview

Visual and tabular view of the Fabric estate, built strictly from metadata captured by the collectors. No customer data is read.

Tenant security posture

Status of the tenant-wide settings evaluated against the checklist.

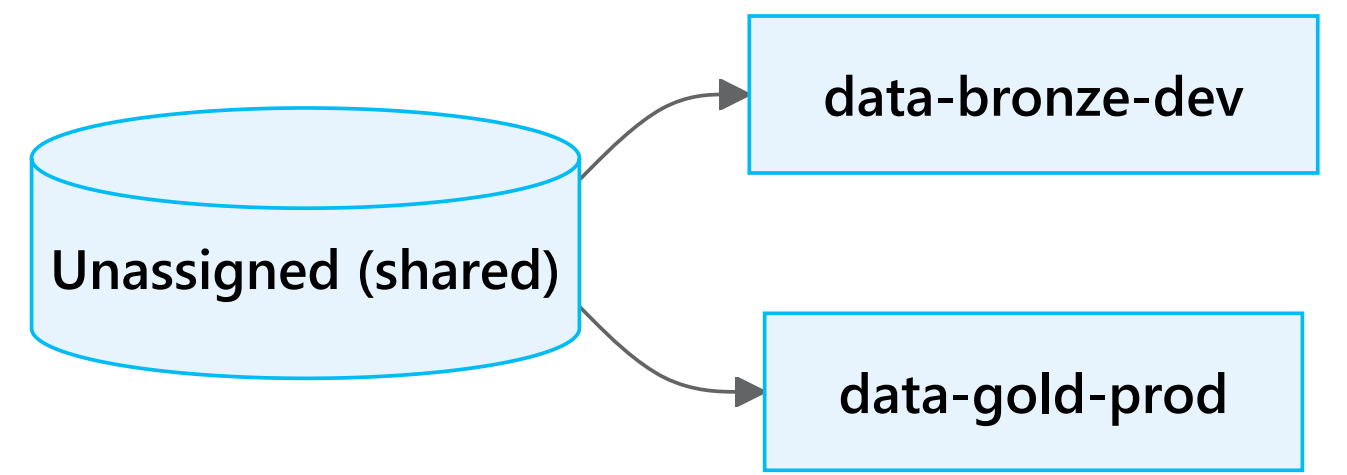


Capacity → Workspace topology

Across 1 capacity bucket(s) and 2 populated workspace(s). Excluded from the architecture view: 1 personal / My workspace entries.

Capacity	Workspaces
Unassigned (shared)	2

Capacity Unassigned (shared) - 2 workspace(s)



Workspace items inventory

Counts of items per workspace returned by the Fabric Scanner API.

Workspace	Lakehouses	Warehouses	Datasets	Reports	Dataflows	Notebooks	Pipelines
data-bronze-dev	1	0	1	1	0	0	0
data-gold-prod	1	0	2	0	0	0	0

Semantic-model storage mode & DirectLake feasibility

3 semantic model(s) in scope: 1 already DirectLake, 2 Import (1 DirectLake candidate(s), 1 blocked by refactor needs). Per-model blockers come from the PERF-012 TMDL audit; **Candidate** means no structural DirectLake blockers were detected in the model definition.

Workspace	Model	Storage mode	DirectLake	Notes
data-bronze-dev	SalesModel	Abf	Blocked	no lakehouse binding, M partitions, calc columns
data-gold-prod	DirectLakeModel	DirectLake	Already DirectLake	—
data-gold-prod	FinanceModel	Abf	Candidate	no structural blockers

Git source-control coverage

1 of 2 workspace(s) are connected to Git source control.

Workspace	Repository	Branch
data-gold-prod	contoso-org/fabric-repo	main

Workspaces without Git (1): data-bronze-dev

Semantic Models — VertiPaq Footprint

In-memory storage-engine statistics for each Import / Direct Lake semantic model: total size, column counts and the most expensive tables and columns — the same numbers DAX Studio's VertiPaq Analyzer reports. Metadata only; no customer rows are read.

VertiPaq analysis did not run in this collection. Run the pipeline inside a Microsoft Fabric notebook to populate model sizes, cardinality and encoding.

DAX Analyzer - Metadata-only static risk

Interpretation: These signals come from measure definitions only. They identify patterns that may be expensive; they are not measured duration, capacity usage, query-plan evidence, or proof that a measure is slow.

- **Coverage:** 3 semantic model(s), 1 measure definition(s), 1 measure(s) with static signals.

Measures with static-risk signals

Workspace	Semantic model	Measure	Risk	Score	Signals	Expression preview
data-bronze-dev	SalesModel	Potentially Expensive Sales	high	80	nested_iterators, crossjoin, whole_table_filter	SUMX(FILTER(SalesFact, SalesFact[Amount] > 0), SUMX(CROSSJOIN(Products, Stores), SalesFact[Amount]))

Detailed Findings

Architecture

2 failing · 1 passing · 1 informational · 0 not applicable · 0 unknown · 0 missing evidence · 4 total

HIGH **FAIL** ARCH-011 — 1 lakehouse-backed Import model(s) may fit Direct Lake

Evidence

- datasetCount: 3
- modeCounts: {"Import": 2, "Direct Lake": 1}
- Directlakecandidates (1): FinanceModel
- Externalimportmodels (1): SalesModel

Examples (3)

name	workspace	storageMode
SalesModel	data-bronze-dev	Import
FinanceModel	data-gold-prod	Import
DirectLakeModel	data-gold-prod	Direct Lake

Recommendation

Evaluate moving Import models over lakehouse data to Direct Lake. Direct Lake reads Delta directly via VertiPaq with no refresh, cutting capacity CU spent on scheduled refresh and reducing data freshness lag.

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview>

MEDIUM **FAIL** ARCH-008 — Personal (My workspace) workspaces in use

Evidence

- personalCount: 1
- Examples (1): My workspace

Recommendation

Migrate any production content out of personal workspaces into shared, capacity-backed workspaces.

Reference: <https://learn.microsoft.com/fabric/get-started/workspaces>

INFO **INFO** ARCH-007 — Empty workspaces (no items)

Evidence

- emptyCount: 1
- Examples (1): My workspace

Recommendation

Archive or repurpose empty workspaces to keep the tenant inventory clean.

Reference: <https://learn.microsoft.com/fabric/governance/governance-overview>

HIGH **PASS** ARCH-013 — Scanner inventory collected completely

Evidence

- workspacesCollected: 3
- workspacesEligible: 3
- batchesTotal: 1

- scoped: no

Recommendation

No action needed — every eligible workspace was scanned.

Reference: <https://learn.microsoft.com/rest/api/power-bi/admin/workspace-info-post-workspace-info>

Performance

11 failing · 1 passing · 1 informational · 0 not applicable · 0 unknown · 0 missing evidence · 13 total

HIGH **FAIL** DAX-001 — 1 measure(s) contain potentially expensive static DAX patterns

Evidence

- metadata only: yes
- measures analyzed: 1
- medium or high risk count: 1
- high risk count: 1

Measures (1)

model_id	model_name	workspace_id	workspace_name
***	SalesModel	***	data-bronze-dev

Recommendation

Review the flagged iterator, cardinality, filtering, materialization, and complexity patterns. Confirm impact with approved runtime tooling before claiming measured performance or cost.

Reference: <https://learn.microsoft.com/power-bi/guidance/dax-avoid-avoid-filter-as-filter-argument>

HIGH **FAIL** PERF-001 — Throttling risk on 1 capacity(ies) (approaching rejection); worst P95 rejection 85%

Evidence

- source: Capacity Metrics App (Usage Summary By Capacities + Items Throttled)
- dataset: {"workspaceId": "****", "name": "Fabric Capacity Metrics"}
- thresholds: {"criticalPct": 100.0, "warnPct": 70.0, "note": "P95 of the Metrics App future-smoothing window; 100% = active throttling (Throttling (s) > 0)."}
- throttledItemsTotal: 0

Offenders (1)

capacityId	health	itemsThrottled	p95BgRejection7d
***	Healthy	0	85

Recommendation

A throttling bucket has reached or is approaching 100% - the point where Fabric actually delays or rejects operations. In the Metrics App, open the named capacity's Throttling page, click the tallest peaks, drill to Timepoint Detail, and sort items by '% of Base capacity' / Total CU (s). Tune the top CU consumers (stagger semantic-model refreshes, reduce query/refresh cost, scope Spark jobs), then enable autoscale or scale-out for the peak windows, or upsize the SKU.

Reference: <https://learn.microsoft.com/fabric/enterprise/throttling>

MEDIUM **FAIL** PERF-003 — Semantic models over size threshold (1 over 2048 MB)

Evidence

- warnThresholdMb: 2048
- criticalThresholdMb: 8192
- modelsWithSizeMetadata: 3

Oversizedmodels (1)

name	workspace	storageMode
SalesModel	data-bronze-dev	Abf

Recommendation

For large Import models, reduce cardinality and unused columns, add aggregations or incremental refresh, and evaluate Direct Lake where source tables live in OneLake Delta.

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview>

MEDIUM **FAIL** PERF-004 — Datasets with high refresh-failure ratio

Evidence

- threshold: 0.2
- count: 1

Datasets (1)

name	workspace	failureRatio	lastStatus
SalesModel	data-bronze-dev	0.5	Failed

Recommendation

Investigate failing datasets — common causes are expired gateway credentials, source schema drift, or out-of-memory in Import models.

Reference: <https://learn.microsoft.com/power-bi/connect-data/refresh-troubleshooting-refresh-scenarios>

MEDIUM **FAIL** PERF-005 — Lakehouse Delta small-file health (1 concern(s))

Evidence

- concernCount: 1

Models (1)

model	concerns
Sales DL	[{"table": "fact_sales", "smallFiles": 9}]

Recommendation

Run OPTIMIZE and enable V-Order for the flagged Delta tables.

Reference: <https://learn.microsoft.com/fabric/data-engineering/lakehouse-table-maintenance>

MEDIUM **FAIL** PERF-006 — Refreshable datasets without a scheduled refresh history

Evidence

- count: 1

Datasets (1)

name	workspace
FinanceModel	data-gold-prod

Recommendation

Configure scheduled refresh (or document the manual-only contract) so data stays current.

Reference: <https://learn.microsoft.com/power-bi/connect-data/refresh-data>

MEDIUM **FAIL** PERF-011 — Capacities running hot (>=70% avg CU 7d) without autoscale

Evidence

Capacities (2)

name	sku	avgCU7d	autoscaleEnabled
Capacity Dev	F2	null	no
Capacity Prod	F64	78	no

Hotwithoutautoscale (1)

name	sku	avgCU7d	autoscaleEnabled
Capacity Prod	F64	78	no

Recommendation

Enable autoscale (or schedule a manual scale-out window) on capacities that regularly exceed 70% average CU - otherwise the next workload increase will throttle.

Reference: <https://learn.microsoft.com/fabric/enterprise/scale-capacity>

MEDIUM **FAIL** PERF-012 — PERF-012: SalesModel — Import-mode model blocks DirectLake migration (no_lakehouse_binding, m_partitions, calculated_columns)

Evidence

- workspace: data-bronze-dev
- workspaceId: ***
- datasetId: ***
- storageMode: Abf
- Direct Lake audit summary: storage partitions are import ; Power Query/M partitions: yes; source connectors: Web.Contents; calculated columns: 1; calculated tables: 0

Direct Lake blockers (3)

Blocker	Why it matters	Evidence
no lakehouse binding	No entityName / DirectLake table binding found; data is materialised via M.	
m partitions	Power Query (M) partitions present; DirectLake disallows M shaping.	Web.Contents
calculated columns	1 calculated column(s) detected; DirectLake-on-Lakehouse disallows them.	1 object(s)

Calculated columns (1)

These columns are defined by DAX expressions in the semantic model. Direct Lake does not support them as model-side calculated columns; move the logic upstream into Delta tables or replace it with measures where appropriate.

Table	Calculated column
SalesFact	Margin

Model table sources (1)

This table shows where each model table appears to come from. **Power Query/M** means the model uses an import query; **calculated table** means the table is produced by a DAX calculated partition. Source hints are metadata-only and may point to SQL objects, files, inline tables, or another model query/expression.

Model table	Partition/source type	Source hint
SalesFact	Power Query/M	
- Metadata coverage note: the analyzer found 0 source-column mapping(s) in the model definition. They are retained in findings_storage_mode.json for traceability, but the PDF focuses on Direct Lake blockers, calculated objects, and table sources.		

Recommendation

DirectLake is not reachable for this model without a refactor: replace M-based partitions with a direct lakehouse-table binding (entityName), push calculated columns/tables into the upstream Delta tables (silver/gold), and re-cast any unsupported data types. Until then, keep the model in Import and ensure a scheduled refresh is configured (PERF-006).

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview>

MEDIUM **FAIL** PERF-012 — PERF-012: DirectLake feasibility — 2 Import-mode model(s) audited, 1 blocked, 1 candidate(s)

Evidence

- auditedModels: 2
- blockedModels: 1
- candidateModels: 1
- directLakeModels: 1
- totalDatasets: 3

Recommendation

Use the per-model PERF-012 findings to plan the storage-mode roadmap. Models flagged 'candidate' can be migrated to DirectLake with little rework; 'blocked' models need the listed blockers cleared first.

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview>

MEDIUM **FAIL** PERF-013 — PERF-013: 0 Direct Lake model(s) pinned to DirectQuery, 1 relying on implicit (automatic) fallback

Evidence

- directLakeModels: 1
- explicitNoFallback: 0

Implicitautomaticfallback (1)

name	workspace
DirectLakeModel	data-gold-prod

Recommendation

Set directLakeBehavior deliberately. Use DirectLakeOnly to fail fast (so guardrail breaches are visible) or Automatic only when a monitored DirectQuery fallback is acceptable. Investigate any model pinned to DirectQueryOnly - it is not getting Direct Lake performance.

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview#fallback-behavior>

MEDIUM **FAIL** PERF-015 — Datasets with no successful refresh in 30 days

Evidence

- thresholdDays: 30
- count: 3

Datasets (3)

name	workspace	lastSuccessfulRefresh
SalesModel	data-bronze-dev	2026-01-10T12:40:00Z
FinanceModel	data-gold-prod	2026-01-13T12:10:00Z
DirectLakeModel	data-gold-prod	2026-01-14T12:05:00Z

Recommendation

Confirm whether these models are still in use; archive or fix the refresh schedule.

Reference: <https://learn.microsoft.com/power-bi/connect-data/refresh-data>

HIGH **INFO** PERF-002 — 7-day average CU% elevated on '***' (78.0%)

Evidence

- source: Capacity Metrics App (Usage Summary By Capacities, 7d/24h/1h)
- dataset: {"workspaceId": "***", "name": "Fabric Capacity Metrics"}

Capacities (1)

capacityId	health	avgCU7d
***	Healthy	78

Recommendation

Average CU% between 70% and 80% is close to throttling threshold. Tune the heaviest workloads now or plan autoscale before the next workload increase.

Reference: <https://learn.microsoft.com/fabric/enterprise/optimize-capacity>

MEDIUM **PASS** DAX-002 — DAX definitions were scanned for 3 semantic model(s)

Evidence

- metadata only: yes
- models scanned: 3
- definition errors: 0
- measures extracted: 1

Recommendation

Keep definition collection enabled so DAX coverage remains current.

Reference: <https://learn.microsoft.com/rest/api/fabric/semanticmodel/items/get-semantic-model-definition>

Governance

2 failing · 1 passing · 1 informational · 0 not applicable · 0 unknown · 0 missing evidence · 4 total

MEDIUM **FAIL** GOV-003 — Sensitivity label coverage

Evidence

- evaluatedItems: 6
- labelledItems: 0
- ratio: 0

Recommendation

Apply sensitivity labels to semantic models, reports, lakehouses, and warehouses; enforce via tenant setting and Purview integration.

Reference: <https://learn.microsoft.com/fabric/governance/information-protection>

MEDIUM **FAIL** GOV-009 — Production workspaces without endorsed semantic models

Evidence

- productionWorkspaces: 1
- minRatio: 0.5
- offenderCount: 1

Examples (1)

name
data-gold-prod

Recommendation

Certify the authoritative semantic models in production workspaces so downstream reports build on governed, trusted data.

Reference: <https://learn.microsoft.com/fabric/governance/endorsement-promote-certify>

LOW INFO GOV-008 — Endorsement (Certified / Promoted) coverage of content

Evidence

- `endorsableItems`: 6
- `endorsedItems`: 0
- `ratio`: 0
- `minRatio`: 0.3

Recommendation

Endorse trusted datasets/reports as Promoted, and the authoritative ones as Certified, so consumers can tell governed content from ad-hoc content.

Reference: <https://learn.microsoft.com/fabric/governance/endorsement-overview>

HIGH PASS GOV-001 — Workspaces with fewer than 2 admins

Evidence

- `evaluatedWorkspaces`: 1
- `minAdmins`: 2
- `underAdminCount`: 0

Recommendation

Assign at least two workspace admins (preferably via a security group) to avoid orphan risk.

Reference: <https://learn.microsoft.com/fabric/get-started/roles-workspaces>

Operational Excellence

0 failing · 1 passing · 0 informational · 0 not applicable · 0 unknown · 0 missing evidence · 1 total

HIGH PASS OPS-001 — Production workspaces covered by a deployment pipeline

Evidence

- `productionWorkspaces`: 1
- `coveredRatio`: 1
- `minRatio`: 0.8
- `uncoveredCount`: 0

Recommendation

Attach every production workspace to a deployment pipeline so content is promoted from validated lower stages instead of edited directly in production.

Reference: <https://learn.microsoft.com/fabric/cicd/deployment-pipelines/intro-to-deployment-pipelines>

Security

4 failing · 1 passing · 1 informational · 0 not applicable · 0 unknown · 0 missing evidence · 6 total

CRITICAL FAIL SEC-001 — Publish to web is disabled or restricted to a security group

Evidence

- `setting name`: PublishToWeb
- `enabled`: yes
- `canSpecifySecurityGroups`: yes

- **reason:** Setting is enabled for the entire organization with no security group scoping.

Recommendation

Tenant setting "Publish to web" is disabled or restricted to a specific security group.

Reference: <https://learn.microsoft.com/power-bi/admin/service-admin-portal-export-sharing#publish-to-web>

HIGH **FAIL** SEC-006 — Misconfigured datasource instances**Evidence**

- **count:** 1

Examples (1)

type	id
Sql	***

Recommendation

Repair gateway bindings, refresh expired OAuth credentials, or remove orphaned datasources.

Reference: <https://learn.microsoft.com/power-bi/connect-data/service-gateway-onprem>

HIGH **FAIL** SEC-008 — Single-member gateway cluster(s) bound to data sources (SPOF)**Evidence**

- **clusterCount:** 1

Singlememberclusters (1)

name	gatewayType	memberCount	datasourceCount
on-prem-cluster	OnPremises	1	2

Recommendation

Add at least one secondary gateway member to every cluster carrying production data sources to remove the single point of failure.

Reference: <https://learn.microsoft.com/data-integration/gateway/service-gateway-high-availability-clusters>

MEDIUM **FAIL** SEC-004 — Workspaces with individual users (not groups) as admins**Evidence**

- **offenderCount:** 2

Examples (2)

workspace
data-bronze-dev
My workspace

Recommendation

Replace individual admin assignments with Entra security groups for lifecycle management.

Reference: <https://learn.microsoft.com/fabric/get-started/roles-workspaces>

HIGH **INFO** SEC-003 — Guest user access setting not returned**Evidence**

- **reason:** Tenant settings did not include AllowGuestUserToAccessSharedContent.

Recommendation

Verify the signed-in user is a Fabric Administrator and re-run the tenant collector.

Reference: <https://learn.microsoft.com/fabric/admin/service-admin-portal-export-sharing>

MEDIUM **PASS** SEC-005 — Workspaces with >10 individual principals

Evidence

- threshold: 10
- count: 0

Recommendation

Consolidate broad access into Entra security groups; remove unused direct assignments.

Reference: <https://learn.microsoft.com/fabric/get-started/roles-workspaces>

Cost

2 failing · 1 passing · 0 informational · 0 not applicable · 0 unknown · 0 missing evidence · 3 total

MEDIUM **FAIL** COST-002 — Reviewer attested Pause/Resume but no Azure automation matched the capacity

Evidence

- reason: CAPACITY_AUTO_PAUSE_CONFIGURED is set, and the Azure ARM scan ran successfully, but no runbook or Logic App content referenced this tenant's Fabric capacities together with suspend/resume verbs. Either the automation lives in a subscription this user cannot read, the runbook content endpoint returned 403, or the attestation is stale.
- subscriptionsScanned: 1
- currentlyPaused: 0

Capacitiesatscan (2)

name	sku
Capacity Dev	F2
Capacity Prod	F64

Recommendation

Either grant the signed-in user Reader on the subscription that hosts the automation account / Logic App and re-run, or unset CAPACITY_AUTO_PAUSE_CONFIGURED if the schedule no longer exists.

Reference: <https://learn.microsoft.com/fabric/enterprise/pause-resume>

MEDIUM **FAIL** COST-005 — Large capacities (F64+) hosting fewer than 5 workspaces

Evidence

- count: 1

Capacities (1)

name	sku	workspaces
Capacity Prod	F64	2

Recommendation

Right-size: either consolidate workspaces onto this capacity or downgrade the SKU.

Reference: <https://learn.microsoft.com/fabric/enterprise/optimize-capacity>

HIGH **PASS** COST-001 — Capacity right-sizing from sustained CU% (0 outside target band)

Evidence

- idleBelowPct: 20
- saturatedAbovePct: 85
- capacitiesEvaluated: 1

Recommendation

Downsize or consolidate chronically idle capacities; optimize workloads, enable autoscale, or upsize chronically saturated capacities.

Reference: <https://learn.microsoft.com/fabric/enterprise/optimize-capacity>

Tenant Settings

1 failing · 1 passing · 0 informational · 0 not applicable · 0 unknown · 0 missing evidence · 2 total

HIGH **FAIL** TENANT-003 — External data sharing is disabled or scoped to a security group

Evidence

- setting name: AllowExternalDataSharing
- enabled: yes
- canSpecifySecurityGroups: yes
- reason: Setting is enabled for the entire organization with no security group scoping.

Recommendation

External sharing tenant settings ("Allow sharing to external users", "Invite external users to your organization") are disabled or scoped to a security group. Tenant-wide external sharing is a major data-exfiltration vector.

Reference: <https://learn.microsoft.com/fabric/admin/service-admin-portal-export-sharing>

HIGH **PASS** TENANT-001 — Users can create Fabric items is scoped to a security group

Evidence

- setting name: CreateFabricItem
- enabled: yes
- canSpecifySecurityGroups: yes
- reason: Enabled but scoped to 1 security group(s).
- Enabledsecuritygroups (1): ***

Recommendation

"Users can create Fabric items" is scoped to a specific security group, not enabled for the entire organization.

Reference: <https://learn.microsoft.com/fabric/admin/fabric-switch>

Notebook Code Review (heuristic)

6 failing · 0 passing · 0 informational · 0 not applicable · 0 unknown · 0 missing evidence · 6 total

HIGH **FAIL** NBCODE-001 — Notebook cells contain potential hard-coded secrets (1 cell(s) across 1 notebook(s))

Evidence

- notebooksScanned: 2
- codeCellsScanned: 2
- matchedCellCount: 1
- matchedNotebookCount: 1
- heuristic: yes
- note: Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Replace literal keys / SAS / tokens / passwords with Key Vault lookups via `notebookutils.credentials.getSecret(...)` or a workspace-managed-identity flow.

Reference: <https://learn.microsoft.com/fabric/data-engineering/microsoft-spark-utilities>

MEDIUM **FAIL** NBCODE-002 — Notebook cells install packages inline (%pip / !pip / %conda) (1 cell(s) across 1 notebook(s))
Evidence

- notebooksScanned: 2
- codeCellsScanned: 2
- matchedCellCount: 1
- matchedNotebookCount: 1
- heuristic: yes
- note: Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Move dependencies to a Fabric environment so installs are baked in once, not repeated on every run, and so library versions are pinned across promotions.

Reference: <https://learn.microsoft.com/fabric/data-engineering/environment-manage-library>

MEDIUM **FAIL** NBCODE-003 — Notebook cells call unbounded .collect() / .toPandas() on Spark DataFrames (1 cell(s) across 1 notebook(s))
Evidence

- notebooksScanned: 2
- codeCellsScanned: 2
- matchedCellCount: 1
- matchedNotebookCount: 1
- heuristic: yes
- note: Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Bare `.collect()/toPandas()` pulls every row to the driver and is a common OOM cause. Bound the result first (`.limit(N)`, `.first()`, `.head(N)`, `.take(N)`, or aggregate to a single row), or persist with `.write.format("delta")` instead of materialising on the driver. Note: Fabric's `display()` is safe — it auto-truncates and is not flagged by this rule.

Reference: <https://learn.microsoft.com/fabric/data-engineering/spark-job-best-practices>

MEDIUM **FAIL** NBCODE-004 — Notebook cells use Databricks-only APIs (1 cell(s) across 1 notebook(s))
Evidence

- notebooksScanned: 2
- codeCellsScanned: 2
- matchedCellCount: 1

- **matchedNotebookCount:** 1
- **heuristic:** yes
- **note:** Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Replace dbutils. *with notebookutils.*, /dbfs paths with abfss:// or attached lakehouse references, and Databricks MLflow URIs with the Fabric MLflow integration.

Reference: <https://learn.microsoft.com/fabric/data-engineering/microsoft-spark-utilities>

LOW **FAIL** **NBCODE-005 — Notebook cells embed hard-coded abfss:// paths or workspace / lakehouse GUIDs (1 cell(s) across 1 notebook(s))**

Evidence

- **notebooksScanned:** 2
- **codeCellsScanned:** 2
- **matchedCellCount:** 1
- **matchedNotebookCount:** 1
- **heuristic:** yes
- **note:** Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Parameterise the lakehouse / path, or rely on the notebook's attached default lakehouse. Hard-coded GUIDs break dev->test->prod promotion via deployment pipelines and Git integration.

Reference: <https://learn.microsoft.com/fabric/onelake/onelake-overview>

LOW **FAIL** **NBCODE-006 — Notebook cells write data in non-Delta formats (parquet / csv / json / orc / avro) (1 cell(s) across 1 notebook(s))**

Evidence

- **notebooksScanned:** 2
- **codeCellsScanned:** 2
- **matchedCellCount:** 1
- **matchedNotebookCount:** 1
- **heuristic:** yes
- **note:** Pattern-based scan — review each match before acting; false positives are possible.

Examples (1)

workspace	notebook	cellIndexes
data-bronze-dev	ProcessSales	[0]

Recommendation

Prefer .format("delta"). Delta gives ACID, time travel, V-Order and is required for Direct Lake. Non-Delta tables silently lose those capabilities in OneLake.

Reference: <https://learn.microsoft.com/fabric/data-engineering/lakehouse-and-delta-tables>

Best Practices

6 failing · 0 passing · 1 informational · 0 not applicable · 0 unknown · 0 missing evidence · 7 total

HIGH **FAIL** BPA-001 — BPA-001: 2 model best-practice violation(s) across 2 model(s)

Evidence

- violations: 2
- Models (1): Sales DL

Recommendation

Clear high-impact BPA rule violations (storage, formatting, perf) in the flagged models.

Reference: <https://learn.microsoft.com/fabric/data-warehouse/semantic-link-labs>

HIGH **FAIL** BPA-002 — BPA-002: 1 report issue(s) across 2 report(s)

Evidence

- issues: 1
- Reports (1): Exec Dash

Recommendation

Fix reports binding to missing fields or with orphaned visuals before publishing.

Reference: <https://learn.microsoft.com/power-bi/create-reports/>

HIGH **FAIL** BPA-003 — BPA-003: 1 Direct Lake fallback reason(s) detected

Evidence

- fallbackReasons: 1
- Models (1): Sales DL

Recommendation

Resolve fallback causes (guardrails, unsupported types) so models stay on Direct Lake.

Reference: <https://learn.microsoft.com/fabric/get-started/direct-lake-overview>

MEDIUM **FAIL** BPA-005 — BPA-005: 1 Delta table health concern(s)

Evidence

- concerns: 1
- Models (1): Sales DL

Recommendation

Run OPTIMIZE/V-Order and consolidate small files for the flagged Delta tables.

Reference: <https://learn.microsoft.com/fabric/data-engineering/delta-optimization-and-v-order>

MEDIUM **FAIL** BPA-006 — BPA-006: 1 unused model object(s)

Evidence

- unused: 1
- Models (1): Sales DL

Recommendation

Remove unused columns/tables/measures to shrink models and refresh time.

Reference: <https://learn.microsoft.com/fabric/data-warehouse/semantic-link-labs>

MEDIUM **FAIL** BPA-007 — BPA-007: 1 capacity(ies) on legacy P-SKU need F-SKU migration

Evidence

- total: 2
- Pskus (1): P1-legacy

Recommendation

Plan migration from P-SKU Premium capacities to Fabric F-SKUs before EOL.

Reference: <https://learn.microsoft.com/fabric/enterprise/capacity-settings>

MEDIUM **INFO** BPA-004 — BPA-004: report BPA evaluated for 2 report(s)

Evidence

- reports: 2
- issues: 1

Recommendation

Address medium report BPA items (visual count, slow visuals, layout) over time.

Reference: <https://learn.microsoft.com/fabric/data-warehouse/semantic-link-labs>

Recommendations & Roadmap

Findings are prioritized by severity and the effort typically required to remediate them. Implement Critical and High items immediately, Medium items in the next review cycle, and Low / Info items opportunistically as part of ongoing operations.

Immediate — Critical / High

Rule	Title	Recommendation
SEC-001	Publish to web is disabled or restricted to a security group	Tenant setting "Publish to web" is disabled or restricted to a specific security group.
ARCH-011	1 lakehouse-backed Import model(s) may fit Direct Lake	Evaluate moving Import models over lakehouse data to Direct Lake. Direct Lake reads Delta directly via VertiPaq with no refresh, cutting capacity CU spent on scheduled refresh and reducing data freshness lag.
BPA-001	BPA-001: 2 model best-practice violation(s) across 2 model(s)	Clear high-impact BPA rule violations (storage, formatting, perf) in the flagged models.
BPA-002	BPA-002: 1 report issue(s) across 2 report(s)	Fix reports binding to missing fields or with orphaned visuals before publishing.
BPA-003	BPA-003: 1 Direct Lake fallback reason(s) detected	Resolve fallback causes (guardrails, unsupported types) so models stay on Direct Lake.
DAX-001	1 measure(s) contain potentially expensive static DAX patterns	Review the flagged iterator, cardinality, filtering, materialization, and complexity patterns. Confirm impact with approved runtime tooling before claiming measured performance or cost.
NBCODE-001	Notebook cells contain potential hard-coded secrets (1 cell(s) across 1 notebook(s))	Replace literal keys / SAS / tokens / passwords with Key Vault lookups via <code>notebookutils.credentials.getSecret(...)</code> or a workspace-managed-identity flow.
PERF-001	Throttling risk on 1 capacity(ies) (approaching rejection); worst P95 rejection 85%	A throttling bucket has reached or is approaching 100% - the point where Fabric actually delays or rejects operations. In the Metrics App, open the named capacity's Throttling page, click the tallest peaks, drill to Timepoint Detail, and sort items by '% of Base capacity' / Total CU (s). Tune the top CU consumers (stagger semantic-model refreshes, reduce query/refresh cost, scope Spark jobs), then enable autoscale or scale-out for the peak windows, or upsize the SKU.
SEC-006	Misconfigured datasource instances	Repair gateway bindings, refresh expired OAuth credentials, or remove orphaned datasources.
SEC-008	Single-member gateway cluster(s) bound to data sources (SPOF)	Add at least one secondary gateway member to every cluster carrying production data sources to remove the single point of failure.
TENANT-003	External data sharing is disabled or scoped to a security group	External sharing tenant settings ("Allow sharing to external users", "Invite external users to your organization") are disabled or scoped to a security group. Tenant-wide external sharing is a major data-exfiltration vector.

Next review cycle — Medium

Rule	Title	Recommendation
ARCH-008	Personal (My workspace) workspaces in use	Migrate any production content out of personal workspaces into shared, capacity-backed workspaces.
BPA-005	BPA-005: 1 Delta table health concern(s)	Run OPTIMIZE/V-Order and consolidate small files for the flagged Delta tables.
BPA-006	BPA-006: 1 unused model object(s)	Remove unused columns/tables/measures to shrink models and refresh time.
BPA-007	BPA-007: 1 capacity(ies) on legacy P-SKU need F-SKU migration	Plan migration from P-SKU Premium capacities to Fabric F-SKUs before EOL.
COST-002	Reviewer attested Pause/Resume but no Azure automation matched the capacity	Either grant the signed-in user Reader on the subscription that hosts the automation account / Logic App and re-run, or unset <code>CAPACITY_AUTO_PAUSE_CONFIGURED</code> if the schedule no longer exists.

Rule	Title	Recommendation
COST-005	Large capacities (F64+) hosting fewer than 5 workspaces	Right-size: either consolidate workspaces onto this capacity or downgrade the SKU.
GOV-003	Sensitivity label coverage	Apply sensitivity labels to semantic models, reports, lakehouses, and warehouses; enforce via tenant setting and Purview integration.
GOV-009	Production workspaces without endorsed semantic models	Certify the authoritative semantic models in production workspaces so downstream reports build on governed, trusted data.
NBCODE-002	Notebook cells install packages inline (%pip / !pip / %conda) (1 cell(s) across 1 notebook(s))	Move dependencies to a Fabric environment so installs are baked in once, not repeated on every run, and so library versions are pinned across promotions.
NBCODE-003	Notebook cells call unbounded .collect() / .toPandas() on Spark DataFrames (1 cell(s) across 1 notebook(s))	Bare .collect()/.toPandas() pulls every row to the driver and is a common OOM cause. Bound the result first (.limit(N), .first(), .head(N), .take(N), or aggregate to a single row), or persist with .write.format("delta") instead of materialising on the driver. Note: Fabric's display() is safe — it auto-truncates and is not flagged by this rule.
NBCODE-004	Notebook cells use Databricks-only APIs (1 cell(s) across 1 notebook(s))	Replace dbutils. with <i>notebookutils.</i> , /dbfs paths with abfss:// or attached lakehouse references, and Databricks MLflow URLs with the Fabric MLflow integration.
PERF-003	Semantic models over size threshold (1 over 2048 MB)	For large Import models, reduce cardinality and unused columns, add aggregations or incremental refresh, and evaluate Direct Lake where source tables live in OneLake Delta.
PERF-004	Datasets with high refresh-failure ratio	Investigate failing datasets — common causes are expired gateway credentials, source schema drift, or out-of-memory in Import models.
PERF-005	Lakehouse Delta small-file health (1 concern(s))	Run OPTIMIZE and enable V-Order for the flagged Delta tables.
PERF-015	Datasets with no successful refresh in 30 days	Confirm whether these models are still in use; archive or fix the refresh schedule.
PERF-006	Refreshable datasets without a scheduled refresh history	Configure scheduled refresh (or document the manual-only contract) so data stays current.
PERF-011	Capacities running hot (>=70% avg CU 7d) without autoscale	Enable autoscale (or schedule a manual scale-out window) on capacities that regularly exceed 70% average CU - otherwise the next workload increase will throttle.
SEC-004	Workspaces with individual users (not groups) as admins	Replace individual admin assignments with Entra security groups for lifecycle management.
PERF-012	PERF-012: SalesModel — Import-mode model blocks DirectLake migration (no_lakehouse_binding, m_partitions, calculated_columns)	DirectLake is not reachable for this model without a refactor: replace M-based partitions with a direct lakehouse-table binding (entityName), push calculated columns/tables into the upstream Delta tables (silver/gold), and recast any unsupported data types. Until then, keep the model in Import and ensure a scheduled refresh is configured (PERF-006).
PERF-012	PERF-012: DirectLake feasibility — 2 Import-mode model(s) audited, 1 blocked, 1 candidate(s)	Use the per-model PERF-012 findings to plan the storage-mode roadmap. Models flagged 'candidate' can be migrated to DirectLake with little rework; 'blocked' models need the listed blockers cleared first.
PERF-013	PERF-013: 0 Direct Lake model(s) pinned to DirectQuery, 1 relying on implicit (automatic) fallback	Set directLakeBehavior deliberately. Use DirectLakeOnly to fail fast (so guardrail breaches are visible) or Automatic only when a monitored DirectQuery fallback is acceptable. Investigate any model pinned to DirectQueryOnly - it is not getting Direct Lake performance.

Backlog — Low / Info

Rule	Title
NBCODE-005	Notebook cells embed hard-coded abfss:// paths or workspace / lakehouse GUIDs (1 cell(s) across 1 notebook(s))
NBCODE-006	Notebook cells write data in non-Delta formats (parquet / csv / json / orc / avro) (1 cell(s) across 1 notebook(s))

Data Safety appendix

This assessment runs strictly against metadata, configuration and metrics surfaced by the Microsoft Fabric / Power BI Admin REST APIs and the Fabric Scanner API. **No customer dataset rows, report visuals, lakehouse tables, notebook contents or pipeline payloads were read.** All collector outputs are stored locally in `output/raw/` and can be deleted after the engagement.